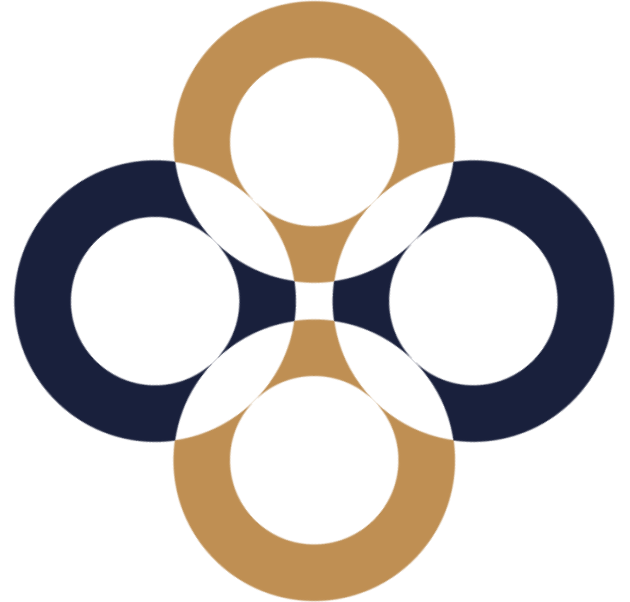


Private Equity

Péter Juhász, PhD, CFA



Sources of financing

Internal

Profit
reinvestment

Unneeded
assets

Debt

Bank loan

Bonds

Equity

Current
investors

New owners

What is private equity?

Specialised investor targeting private companies, willing to obtain control over the firms and exit within some time specified in advance.

How is private equity different?

Projects are considered on a portfolio basis not individually.

Successful projects compensate for capital and return lost in failure projects.

Investor provides knowledge next to capital.

The exit is pre-planned.

Benefits

Management
knowhow

Strategic
view

Efficiency
increase

Synergies

M&A
opportunities

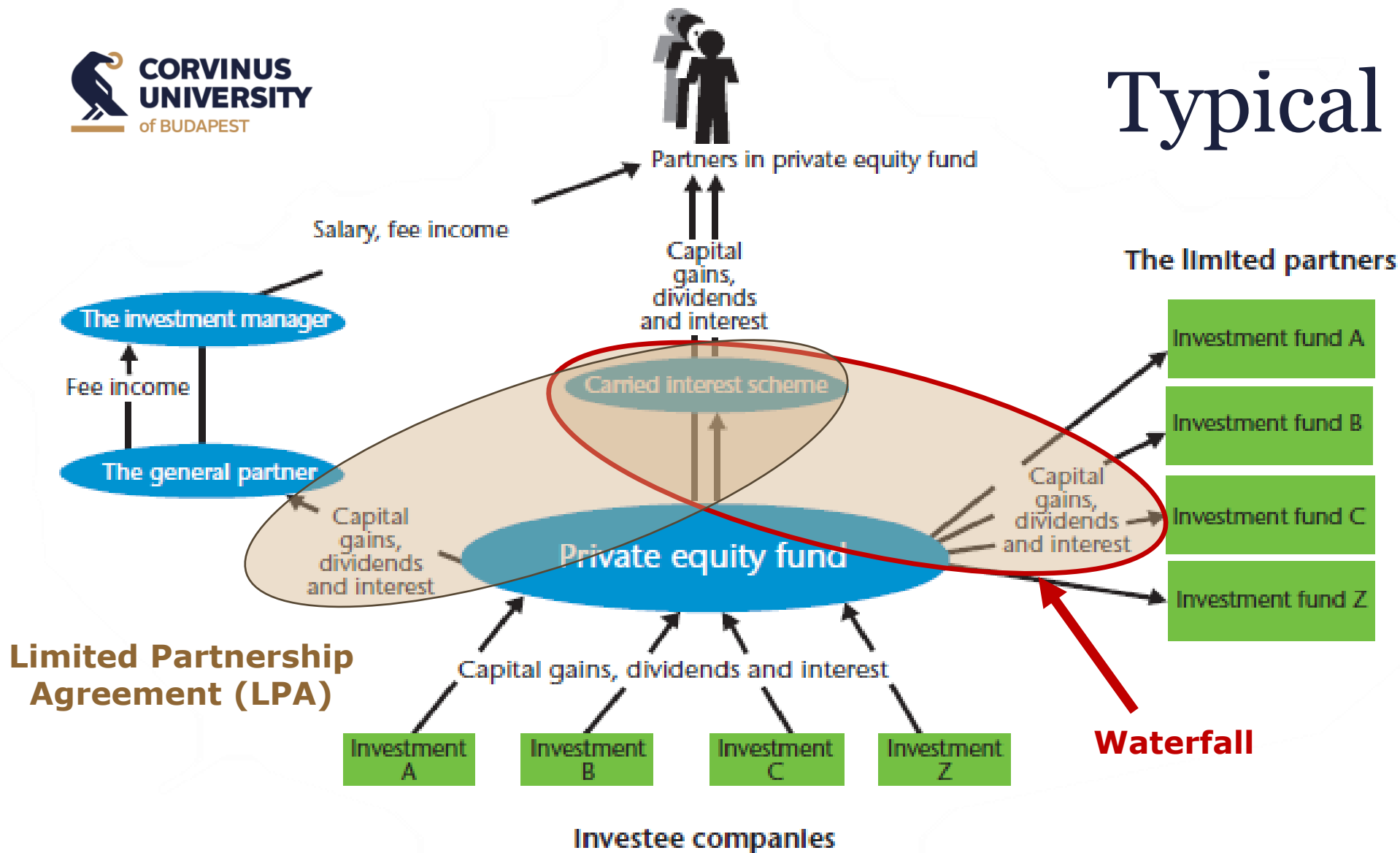
Limit on
agency
problem

Value based
compensation

Where does it come from?



Typical structure



Forms of private equity

Direct

Investor
owns the
shares
directly

Indirect

Investor and
fund
manager
are different
entities

Typical investors

	Direct	Indirect
Individuals	✓	✓
Companies	✓	✓
Banks		✓
Insurance companies		✓
Investment funds		✓
Pension funds		✓
Governmental institutions		✓

Investment policy



Written statement

Specifies target sectors, investment size, selection and decision processes

Investment committee

Investors' council

Categorisation

Sector focus

Sector knowledge

High exposure to a specific sector

Duration of investment and cycle of sector to be harmonised

Sectors before expansion or consolidation

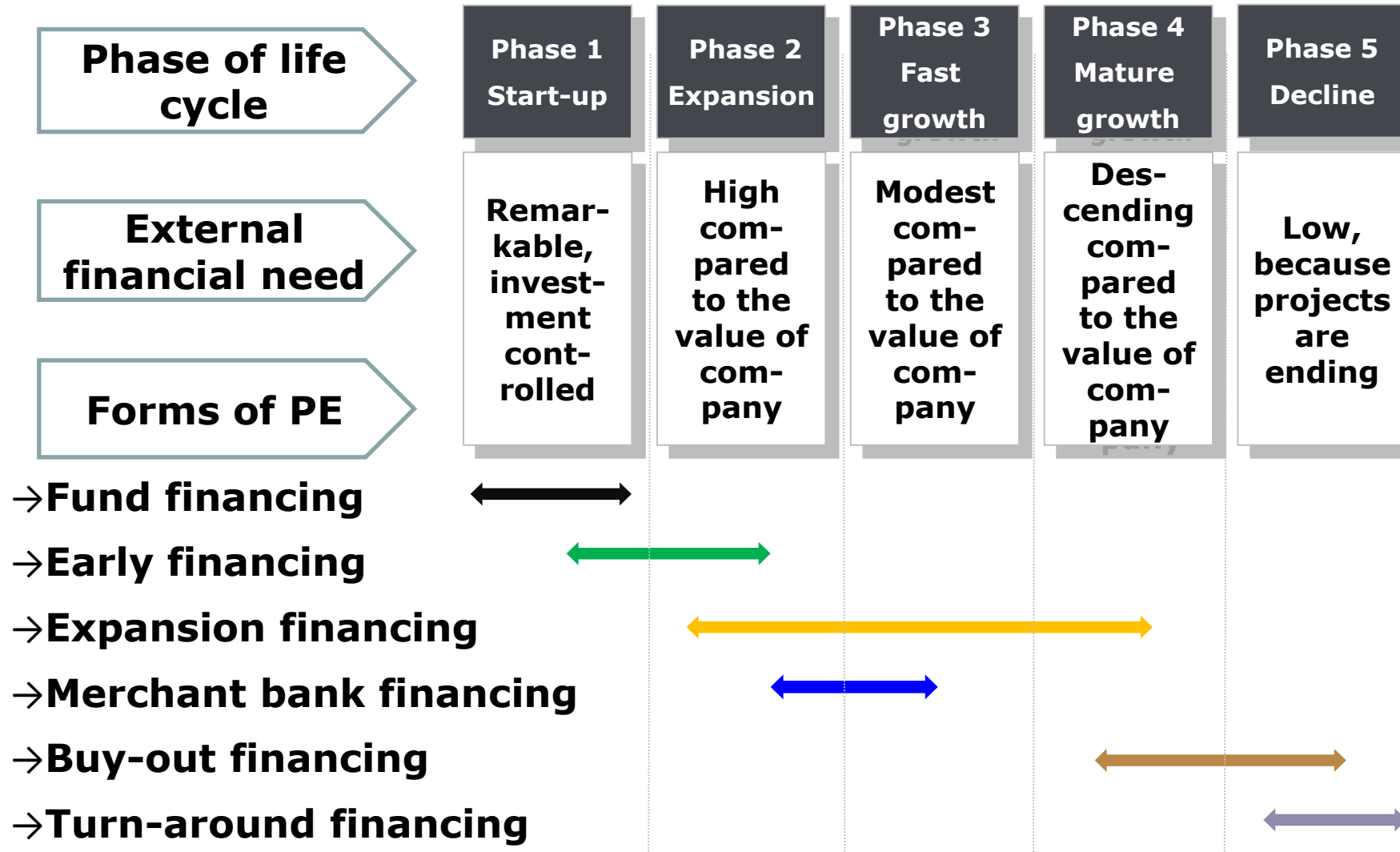
Region focus

Countries in similar economic cycle

Size focus

1/5/20/100/500 M euros

Life cycle focus

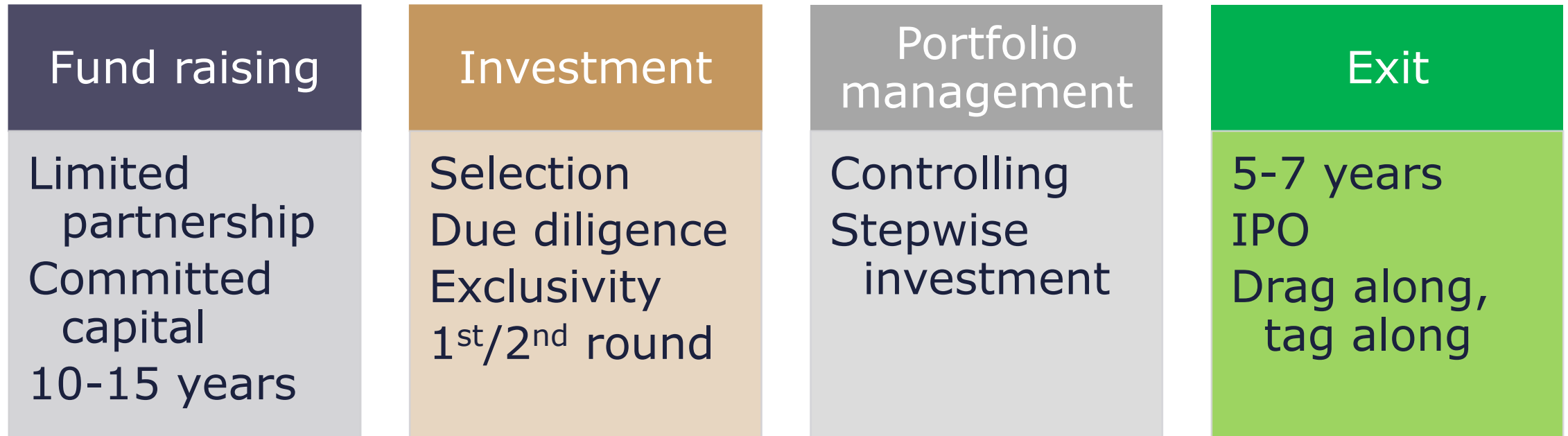


Other categorisations

Leverage

Acquired stake

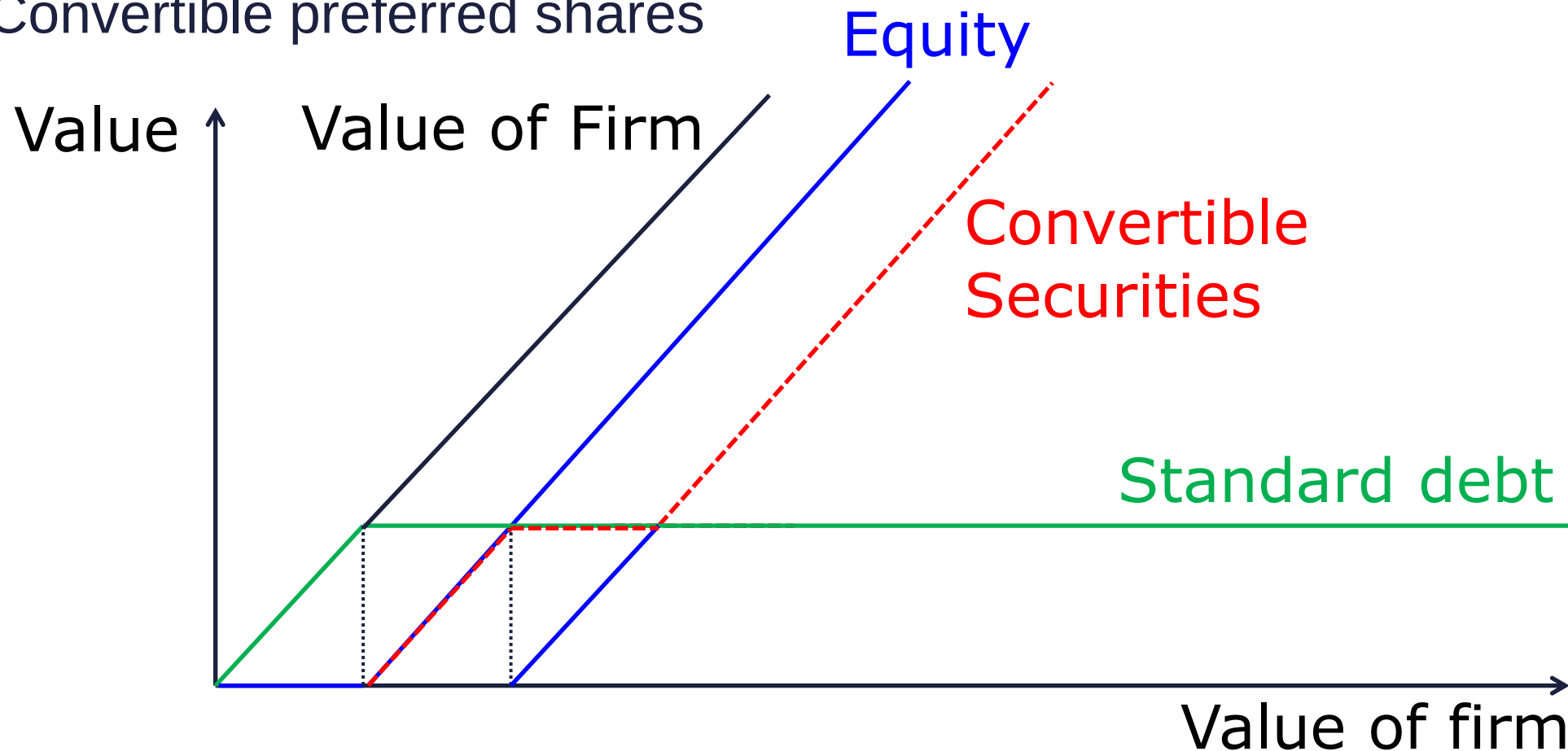
Steps of PE investments



Convertible securities

Convertible loans, bonds + warrants

Convertible preferred shares



Venture capital

Private Equity that invests in start-ups and early-stage deals.

Capital for Start-ups

FFF

Family, Friends, and Fools

Business Angels

Wealthy individuals who invest in start-ups and early-stage deals.

Equity crowdfunding

A group of individuals who invest in start-ups and early-stage deals.

Venture Capital

Private Equity that invest in start-ups and early-stage deals.

Types of Venture capital

Early-stage

Seed, start-up, first-stage

Expansion

Later stage: Acquisition, Buy-out



Dark side of private equity

Growth pressure

Efficiency pressure

Exit pressure

Ethical issues

Conflicts of interest	Buyout/exit not always in the interest of all owners
Vulnerability of employees	Restructuring may mean lay-offs
Capital markets	Public-to-private deals, Insider info
Tax avoidance	PE firms and management usually very good at tax optimising

Required return for VC

At a VC fund with an average investment horizon of 4 years 70% of similar projects fail. In case of default, 110% of initial investment can be collected at the end. Cost of financing is 20%.

Required return for projects surviving:

$$70\% * 110\% + 30\% * (1+r)^4 = 1.2^4$$

$$30\% * (1+r)^4 = 1.3036$$

$$\mathbf{r = 44.38\%}$$

Drivers of investment success

Better
selection
(specialisation)

Close control

Professional
assistance

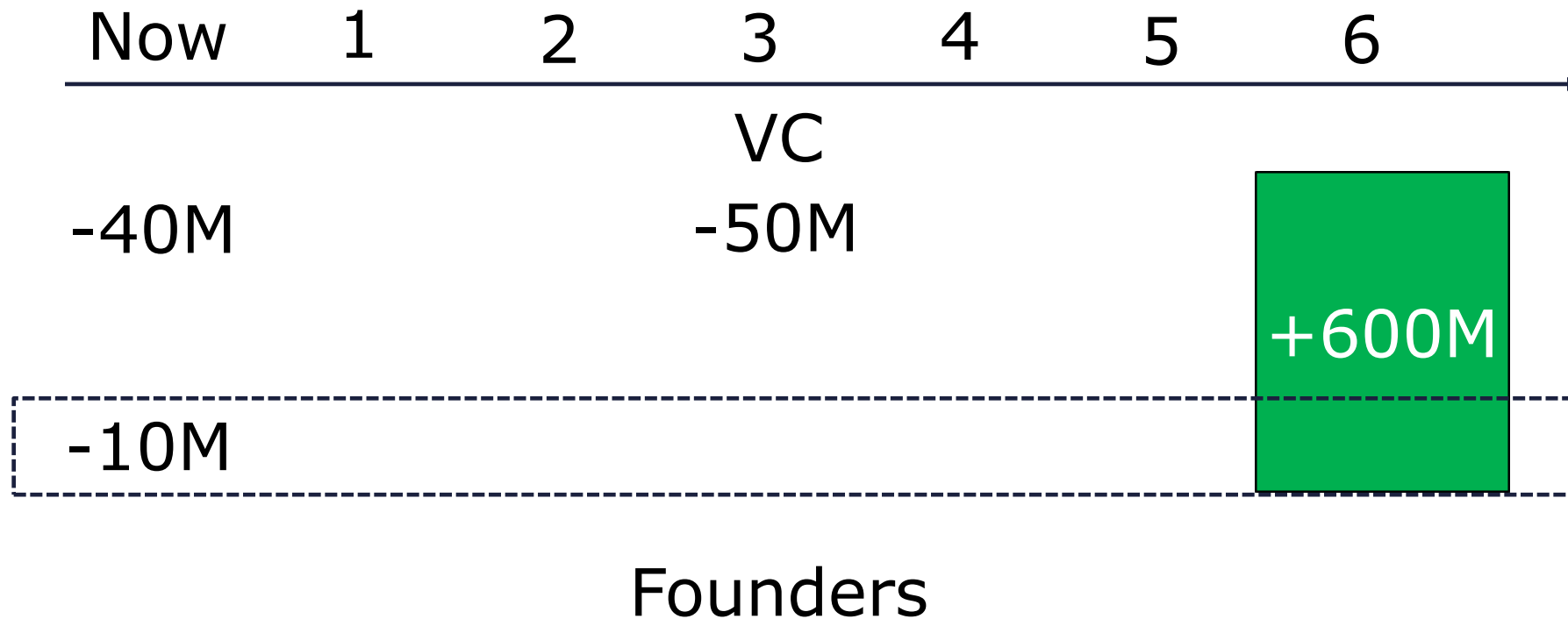
Covenants
and collaterals

Fair price and stake 1.

A firm with a capital of 10 M needs an immediate investment of 40 M. In case of success after 3 years another 50 M will be needed and after 6 years the total of equity may be sold for 600 M.

What stake should VC receive for the equity increases, if required rate of return is 40%?

Investment plan



Fair price and stake 2.

Equity value (E) in 6 years: 600.00 M

In 3 years, after capital: $600.00/1.4^3=218.66$

In 3 years, before capital:

$$218.66-50.00=168.66$$

Stake for investment of 50 M in 3 years:

$$50.00/218.66=22.87\%$$

Calculations

Now

1

2

3

4

5

6

Post-Money 219M

VC -50M

Pre-Money 169M

23%

+600M

VC -40M

Founders -10M

Fair price and stake 3.

In 3 years, before capital: $218.66 - 50.00 = 168.66$

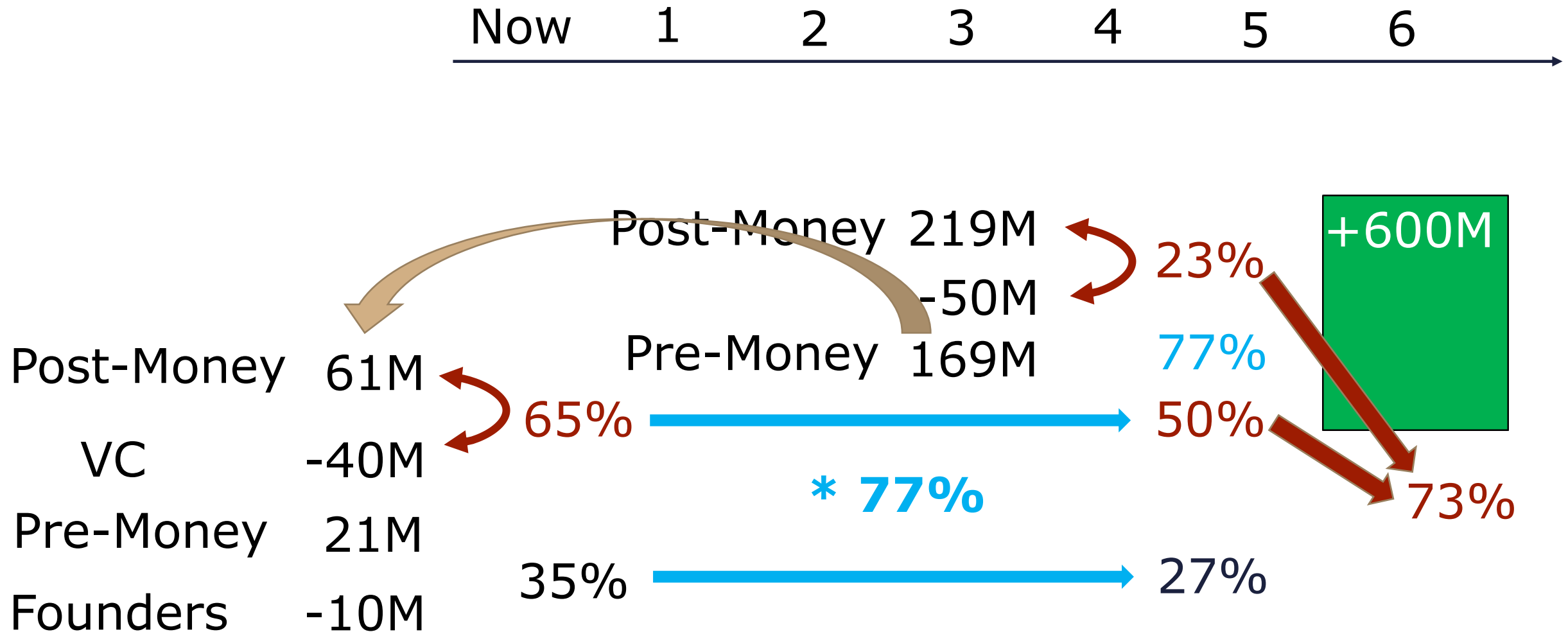
Today, after capital: $168.66 / 1.4^3 = 61.46$

Today, before capital: $61.46 - 40.00 = 21.46$

Stake for current investment:

$$40.00 / 61.46 = 65.08\%$$

Calculations



Fair price and stake 4.

	VC	Founders
Now	0.00%	100.00%
After investment	65.08%	34.92%
In 3 years, after investment	73.07%	26.93%

Stake for capital in year 3: 22.87%

VC total:

$$65.08\% * (1 - 22.87\%) + 22.87\% = 73.07\%$$

Stake of founders:

$$34.92\% * (1 - 22.87\%) = 26.93\%$$

Fair price and stake 5.

Return to founders:

$$600.00 * 26.93\% = 161.58$$

$$161.58 / 21.46 = (1+r)^6$$

$$\mathbf{r = 40.00\%}$$

Measuring value created

EVA:
 $IC \cdot (ROIC - WACC)$

Residual income:
 $PAT - St \cdot r_E$

ΔMVA :
 $(V_1 - IC_1) - (V_0 - IC_0)$

SVA:
 $\Delta E \pm CF$

CFROI:
IRR on FCFE

TSR:
$$\frac{(P_1 - P_0) + Div_1}{P_0}$$

Balanced
Scorecard

The secret of successful VBM

Top
management
support

Including
operational
decision

Ability
development

Practice focus

Individual KPIs

Provide data
access

Standardised
format for
reports and
analysis

Bonuses linked
to value creation

Obligatory
value-based
justification of
resource needs

Source:
McKinsey, 1994

**Thank you
for your attention!**

